1. Background

Customer churn, the phenomenon in which customers discontinue their relationship with a company, represents a major challenge in the retail banking industry. When customers close their accounts or stop using banking services, financial institutions experience revenue loss, reduced cross-selling opportunities, and increased marketing costs required to acquire new customers. Retaining existing customers is generally more cost-effective than acquiring new ones, making churn prediction a critical business objective.

In this project, we aimed to predict customer churn using machine learning techniques based on demographic, financial, and behavioral attributes. The dataset includes customer information such as age, balance, number of products, credit score, tenure, and activity status.

By analyzing the factors associated with churn and comparing multiple classification models, we sought to identify the most important drivers of customer attrition and provide actionable insights that can help banks proactively retain high-risk customers.

2. Research Question

Our research question was:

What factors contribute to customer churn in retail banking, and how accurately can machine learning models predict whether a customer will leave the bank?

To address this question, we explored two modeling approaches:

1. **All Features Approach** – Using the full set of available demographic and financial features in the dataset.

2. **Selected Features Approach** – Using a subset of features selected based on correlation analysis and feature importance, focusing on the variables most strongly associated with churn.

These approaches allowed us to evaluate how feature selection impacts model performance and to better understand which variables are most influential in predicting customer churn.

---

3. Methods

In this section, we describe the methodology used to prepare the dataset, perform exploratory data analysis (EDA), train multiple machine learning models, and evaluate their performance.

3.1 Data Overview

We began by loading a dataset containing customer information from a retail bank. The dataset includes demographic, financial, and behavioral attributes for each customer. Our target variable is **Exited**, which indicates whether a customer left the bank (1) or remained (0).

The dataset consists of:

- Customer demographic features (e.g., Age, Gender, Geography)
- Financial attributes (e.g., Balance, CreditScore, EstimatedSalary)
- Behavioral indicators (e.g., NumOfProducts, IsActiveMember, HasCrCard)
- The target variable: **Exited**

Dataset Structure

The dataset contains a combination of numerical and categorical variables, including:

- Numerical features such as Age, CreditScore, Balance, Tenure, EstimatedSalary, NumOfProducts
- Categorical features such as Geography and Gender
- Target variable: Exited (1 = churned, 0 = retained)

Missing Values

A detailed analysis of missing values showed that the dataset contains **no missing values**. All columns were fully populated and required no imputation before modeling.

Dataset Shape:

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	1	15634602	Hargrave	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	3	15619304	Onio	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
3	4	15701354	Boni	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	5	15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0

Dataset Info:

```
Data columns (total 14 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   RowNumber        10000 non-null  int64
 1   CustomerId       10000 non-null  int64
 2   Surname          10000 non-null  object
 3   CreditScore      10000 non-null  int64
 4   Geography        10000 non-null  object
 5   Gender           10000 non-null  object
 6   Age              10000 non-null  int64
 7   Tenure           10000 non-null  int64
 8   Balance          10000 non-null  float64
 9   NumOfProducts    10000 non-null  int64
 10  HasCrCard        10000 non-null  int64
 11  IsActiveMember   10000 non-null  int64
 12  EstimatedSalary  10000 non-null  float64
 13  Exited           10000 non-null  int64
```

Missing Values:

```
Missing values per column:
```

	0
CreditScore	0
Geography	0
Gender	0
Age	0
Tenure	0
Balance	0
NumOfProducts	0
HasCrCard	0
IsActiveMember	0
EstimatedSalary	0
Exited	0

3.2 Exploratory Data Analysis (EDA)

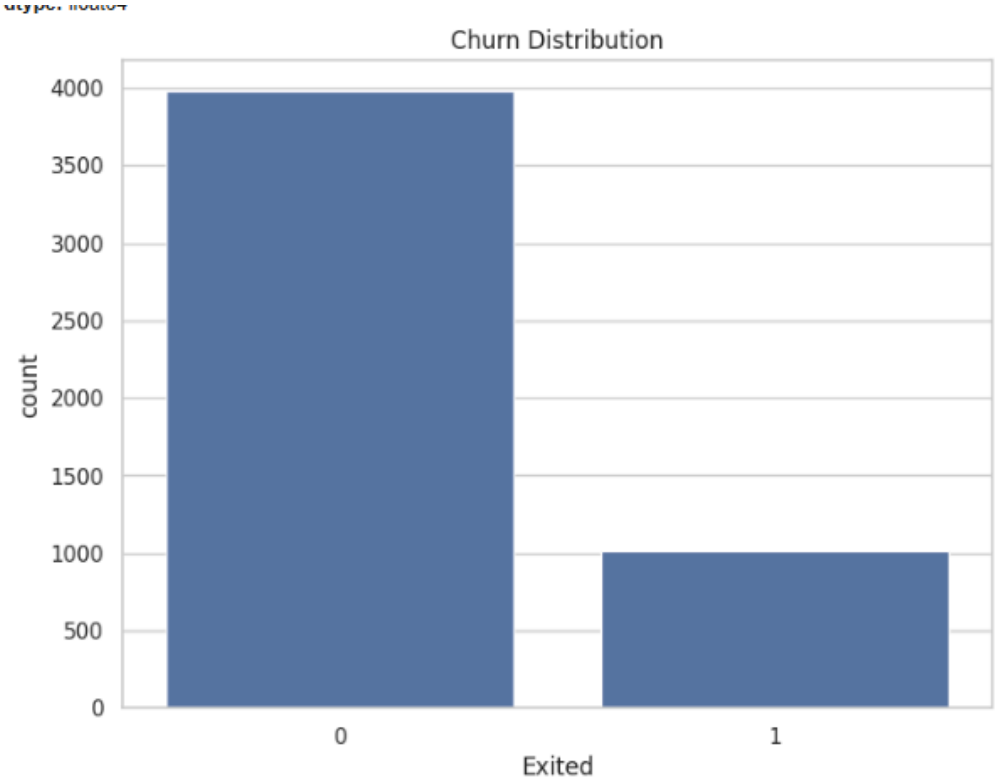
We conducted exploratory data analysis (EDA) to better understand the distribution of features and their relationship with the target variable (**Exited**). The goal of this analysis was to identify potential patterns, class imbalance, and key variables that may influence customer churn.

Below are the main visualizations and findings.

1️⃣ Churn Distribution

The first step was to analyze the distribution of the target variable. The dataset is imbalanced, with significantly more customers who remained with the bank than customers who churned.

This class imbalance suggests that evaluation metrics such as F1-score and ROC-AUC are more appropriate than accuracy alone.



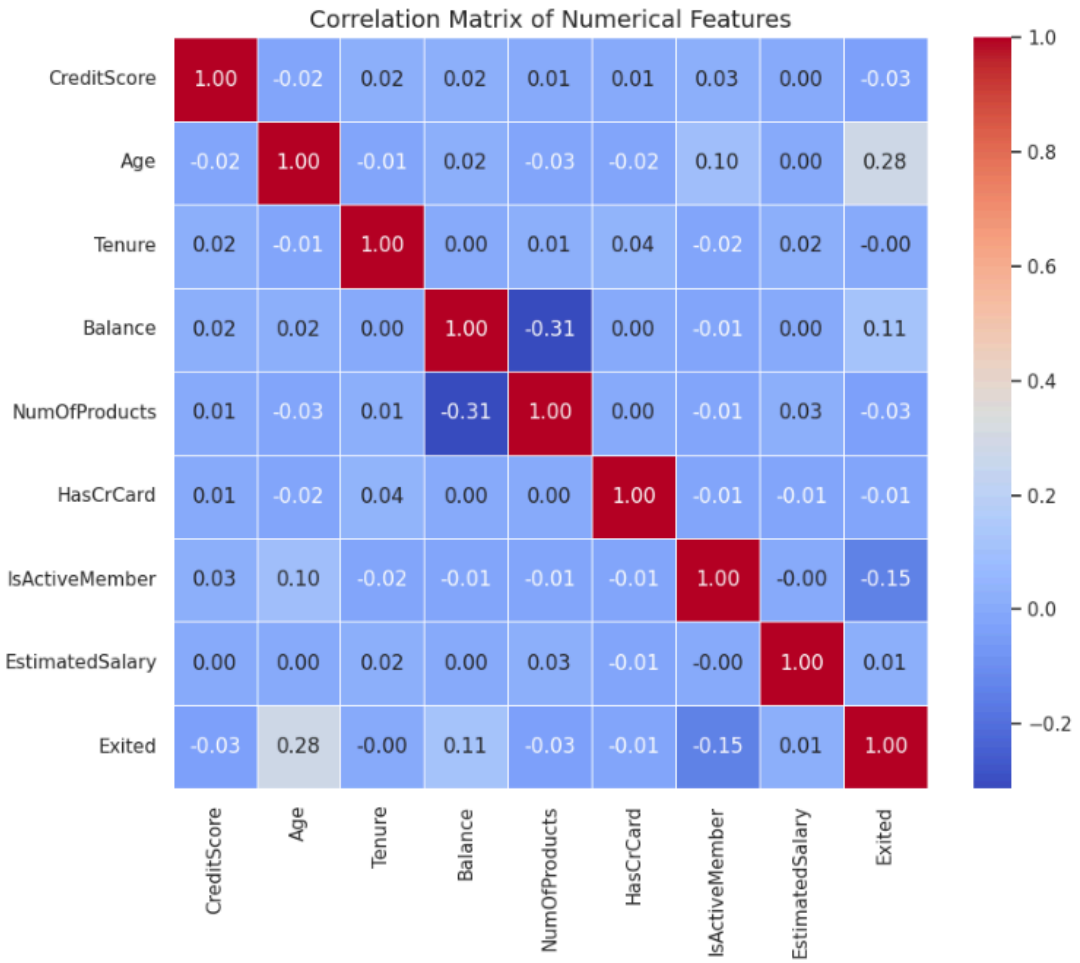
2 Correlation Analysis

To examine relationships between numerical features and churn, we generated a correlation heatmap. This visualization helps identify variables that are strongly associated with the target variable and with each other.

Key variables showing noticeable correlation with churn include:

- Age
- Balance
- IsActiveMember
- NumOfProducts

...

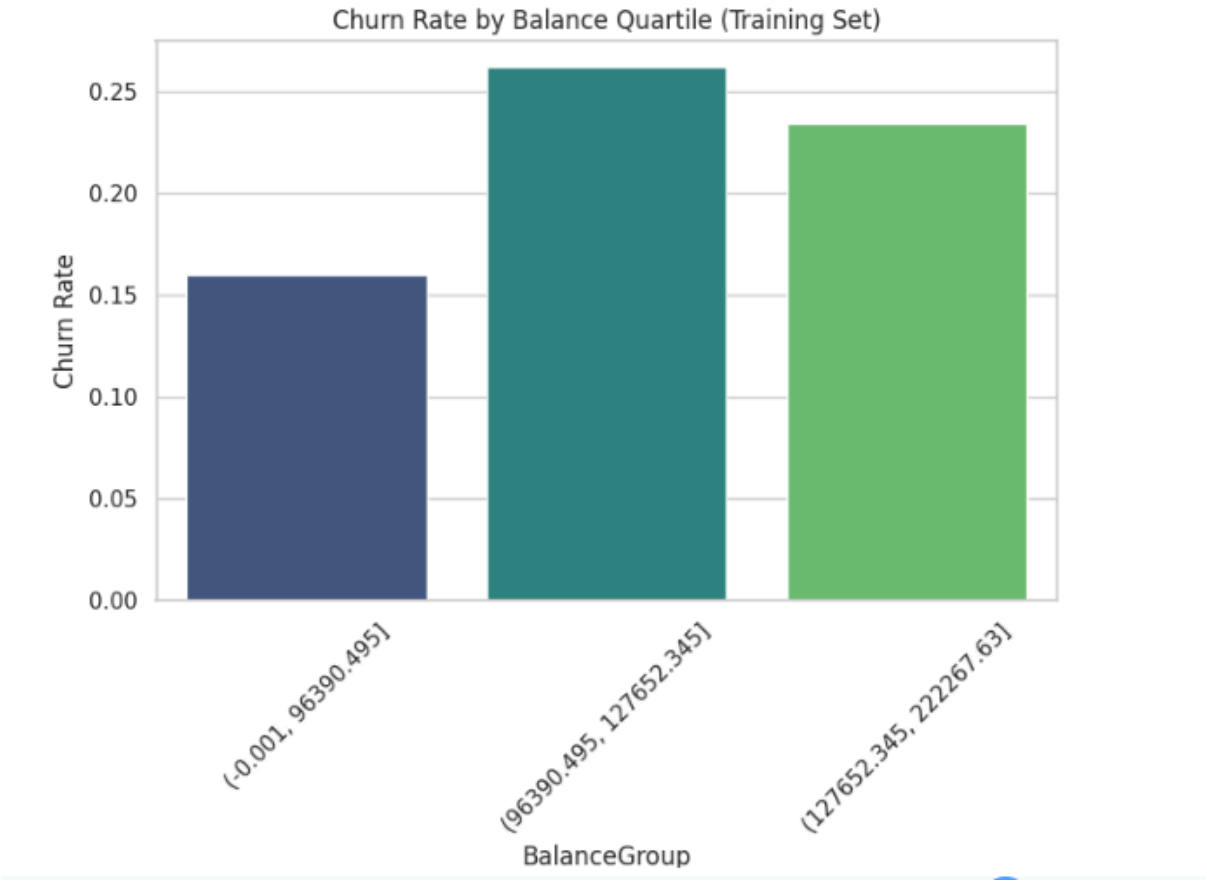


3️⃣ Churn by Balance / Activity / Products

To further investigate churn behavior, we examined how churn varies across key features.

a) Balance

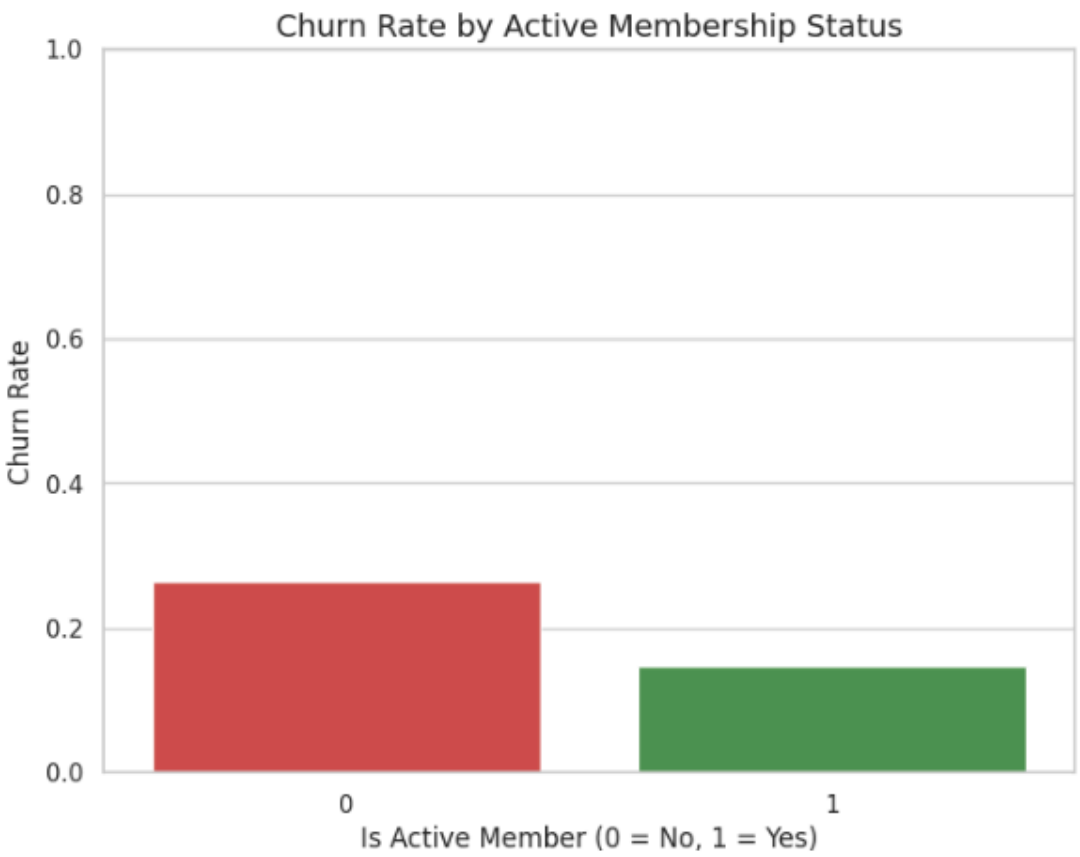
Customers with higher balances showed different churn patterns compared to customers with lower balances.



b) IsActiveMember

Active members were significantly less likely to churn compared to inactive members.

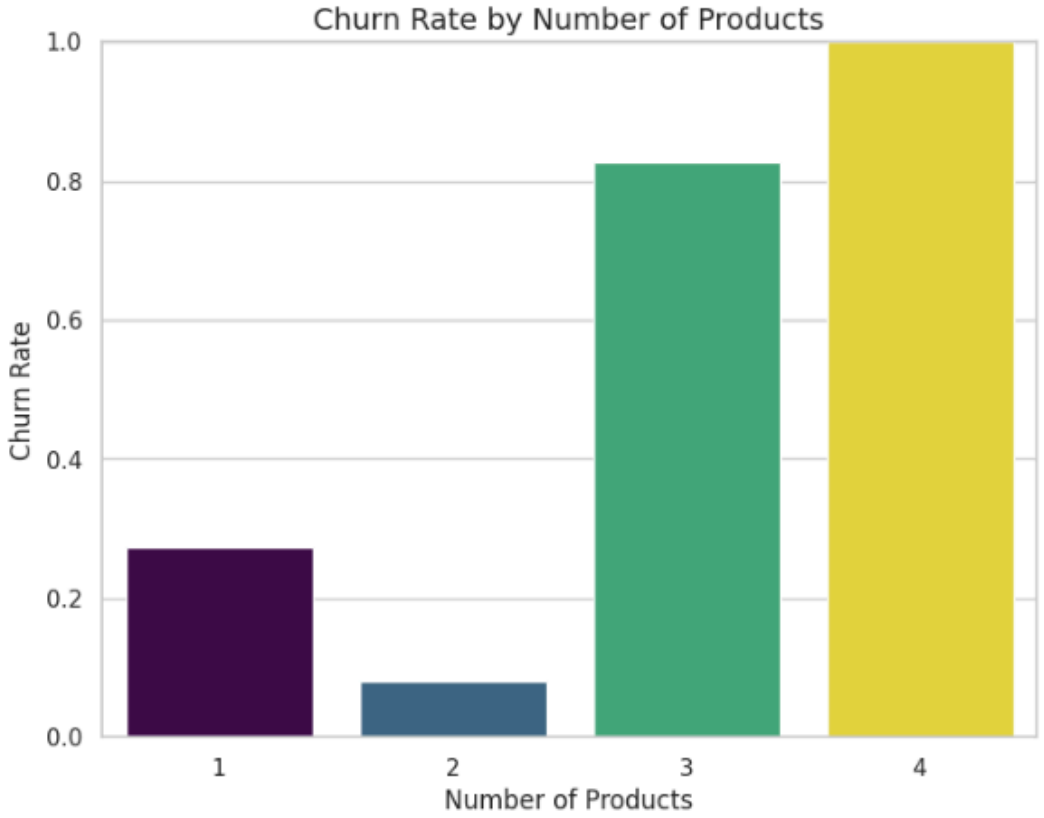
...



c) Number of Products

Churn patterns varied across customers with different numbers of bank products.

...



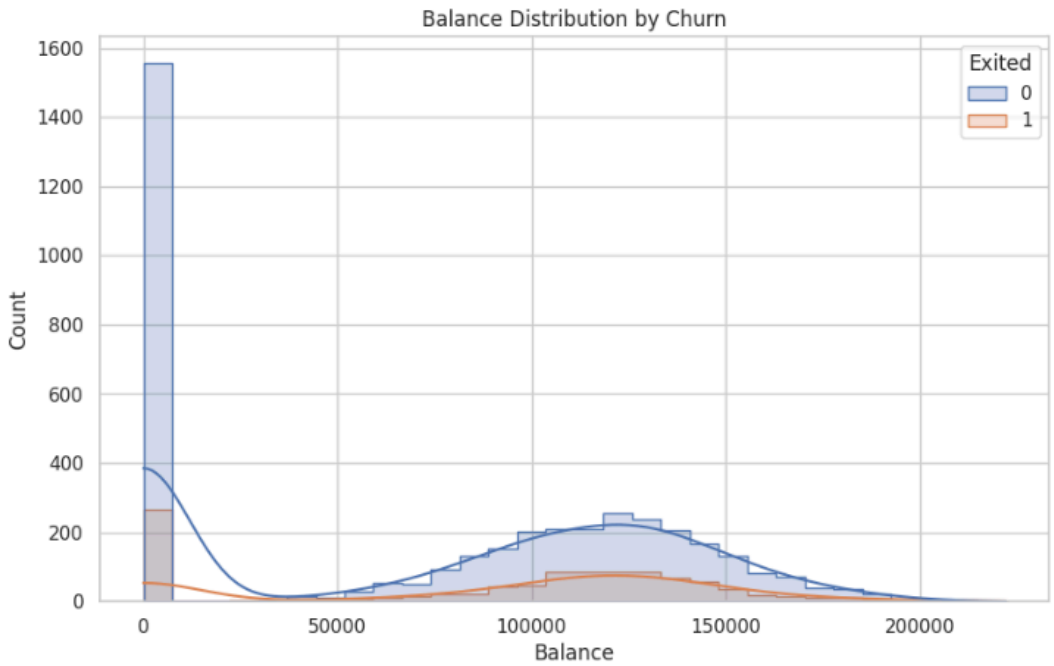
4. Balance Distribution by Churn

We analyzed how customer account balance is distributed among customers who churned versus those who remained with the bank.

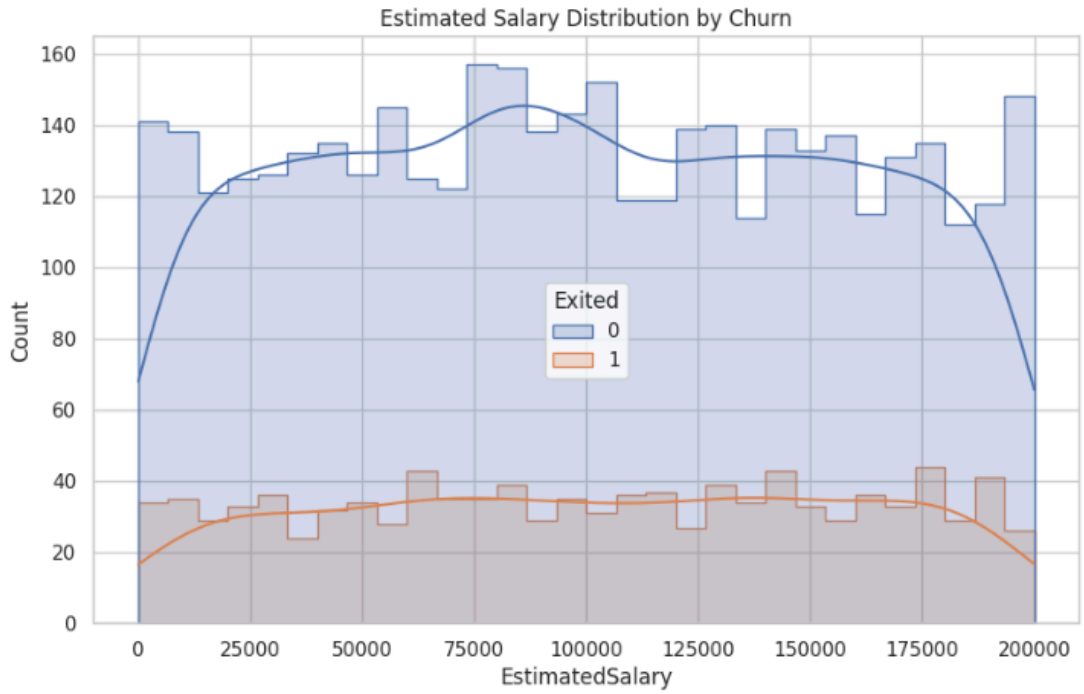
The distribution shows noticeable differences between churned and retained customers. Customers with higher balances tend to exhibit different churn behavior compared to customers with lower balances.

This suggests that account balance may play an important role in predicting customer churn.

...



...



3.3 Data Preprocessing and Feature Engineering

To prepare the data for machine learning models, several preprocessing steps were applied to ensure data quality and prevent data leakage.

---

1. Handling Categorical Variables

The dataset included categorical variables such as:

- Geography
- Gender

These variables were converted into numerical representations using encoding techniques to allow their use in machine learning models.

	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
0	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
3	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0

2. Removing Irrelevant Features

Certain columns were removed because they did not provide predictive value:

- RowNumber
- CustomerId
- Surname

These variables are identifiers and do not contain meaningful information for churn prediction.

	CreditScore	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Geography_Germany	Geography_Spain	Gender_Male
0	619	42	2	0.00	1	1	1	101348.88	False	False	False
1	608	41	1	83807.86	1	0	1	112542.58	False	True	False
2	502	42	8	159660.80	3	1	0	113931.57	False	False	False
3	699	39	1	0.00	2	0	0	93826.63	False	False	False
4	850	43	2	125510.82	1	1	1	79084.10	False	True	False

3. Train / Validation / Test Split

To prevent data leakage and ensure proper evaluation, the dataset was split into:

- Training set
- Validation set

- Test set

Stratified splitting was applied to preserve the original class distribution across the splits.

```
Train: (5000, 11) Churn: 0.2038
Val:   (3000, 11) Churn: 0.20366666666666666
Test:  (2000, 11) Churn: 0.2035
```

4. Feature Scaling

To ensure that numerical variables with different ranges (e.g., Balance and CreditScore) do not bias the models, we applied StandardScaler to the training data. The scaler was fitted only on the training set and then applied to the validation and test sets.

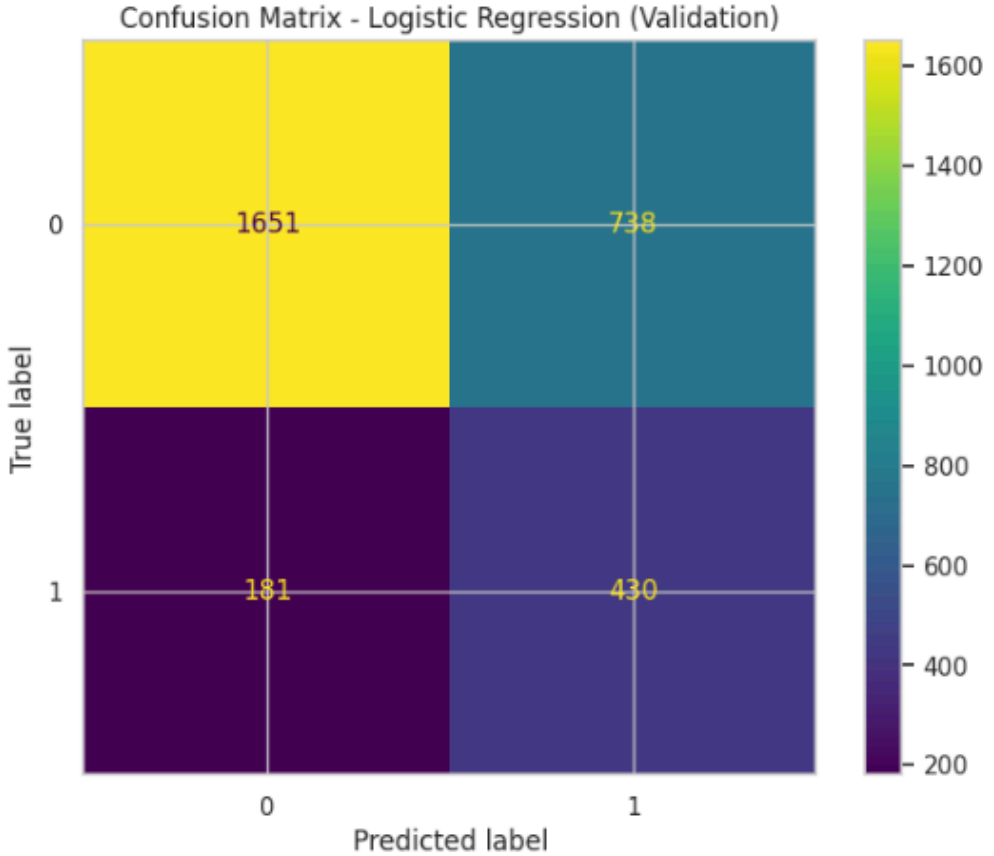
3.5 Model Evaluation

To assess the performance of the models in predicting customer churn (Exited = 1), we used the following evaluation metrics:

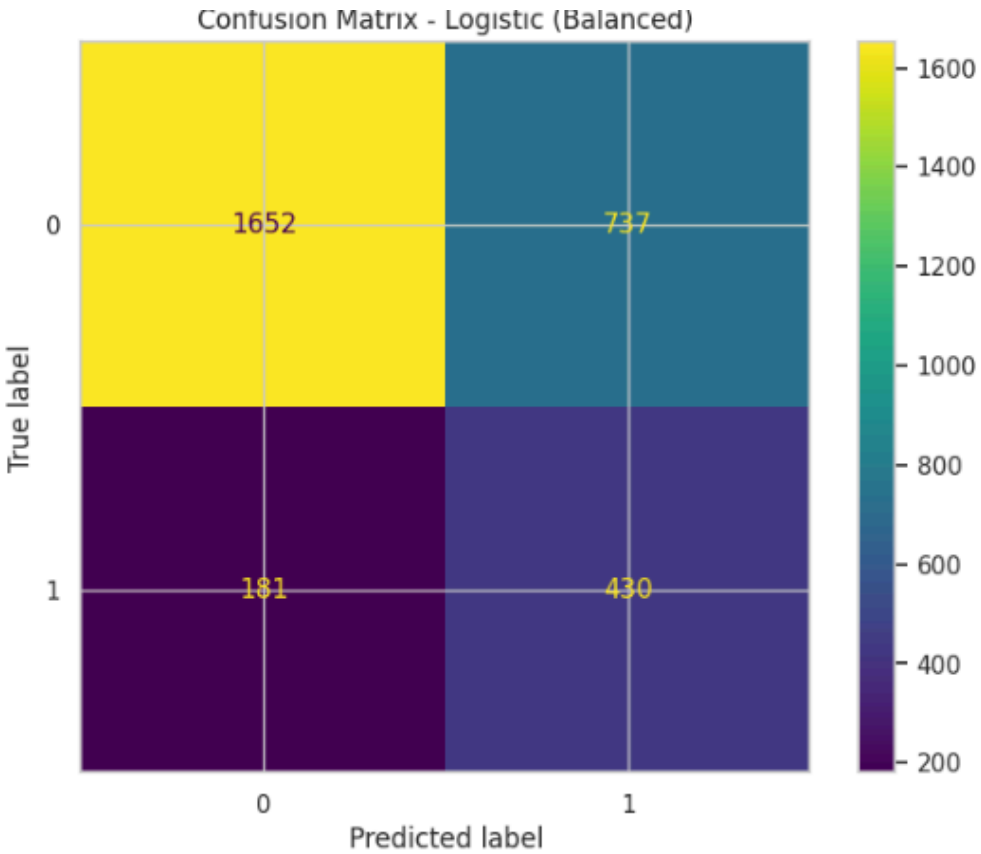
- **Accuracy** – Overall classification accuracy.
- **Precision, Recall, F1-Score** – Performance metrics for each class (Exited = 1 represents customers who left the bank).
- **ROC-AUC** – Measures the model's ability to distinguish between churned and non-churned customers.
- **Confusion Matrix** – A visual representation of the model's predictions versus actual values.

Since the dataset is imbalanced (approximately 20% churn), special attention was given to the **Recall score for churned customers**, as correctly identifying customers who are likely to leave is the primary business objective.

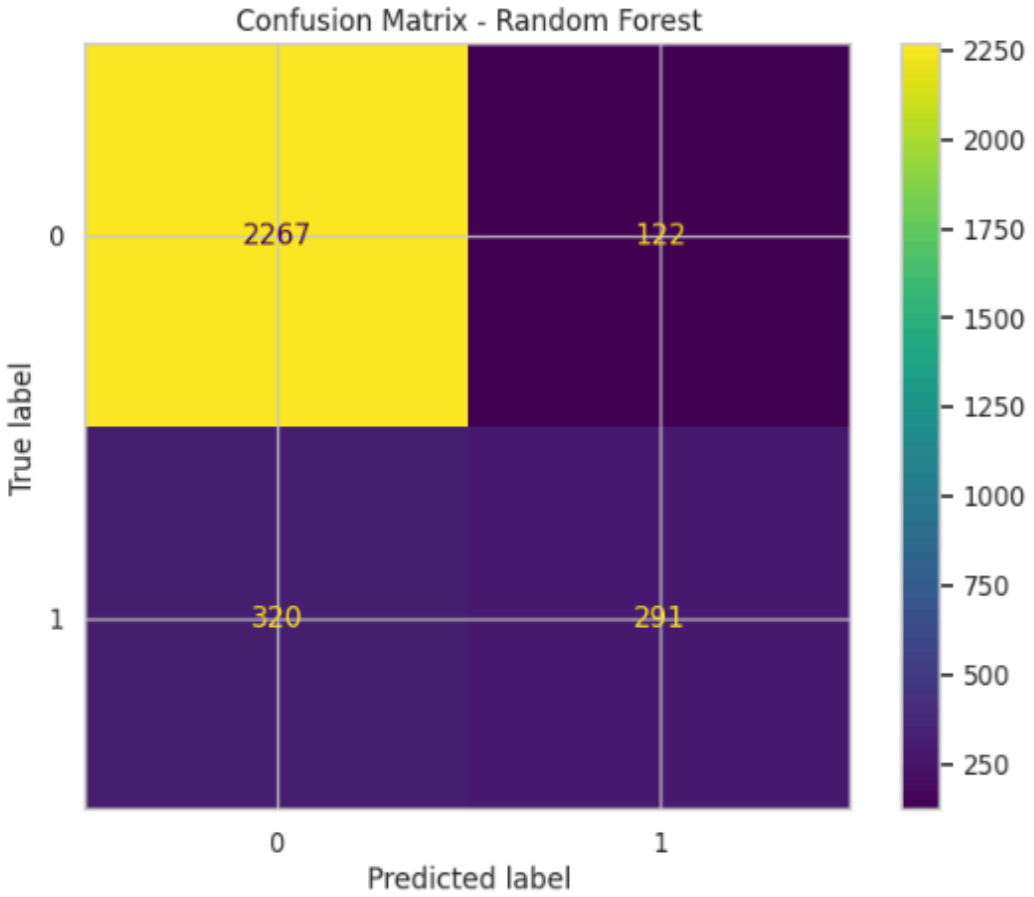
Logistic Regression – Validation Set



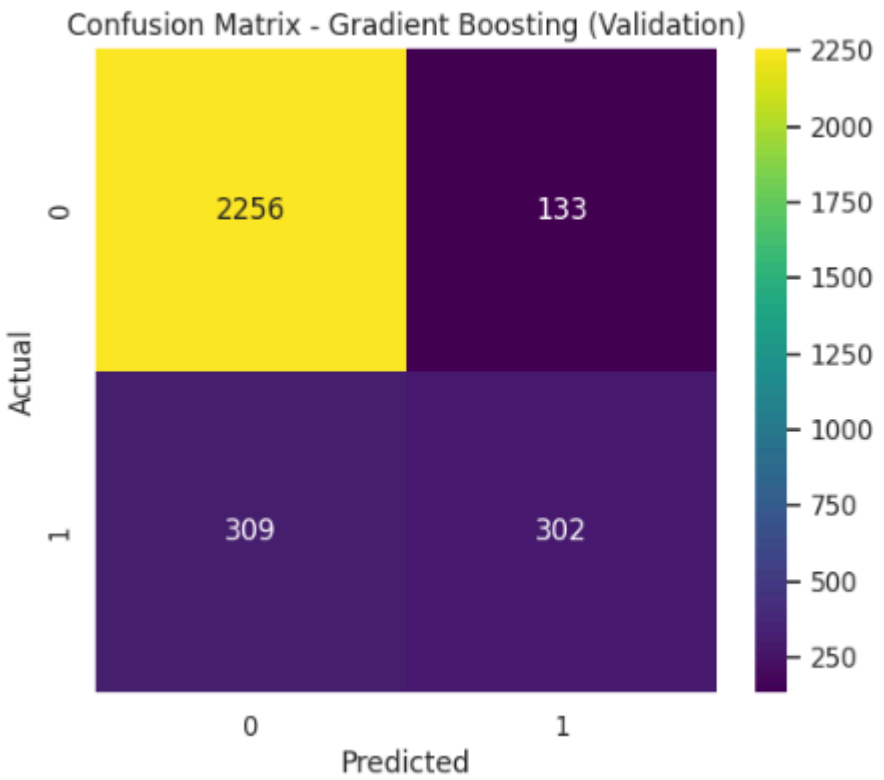
Logistic Regression (Balanced)



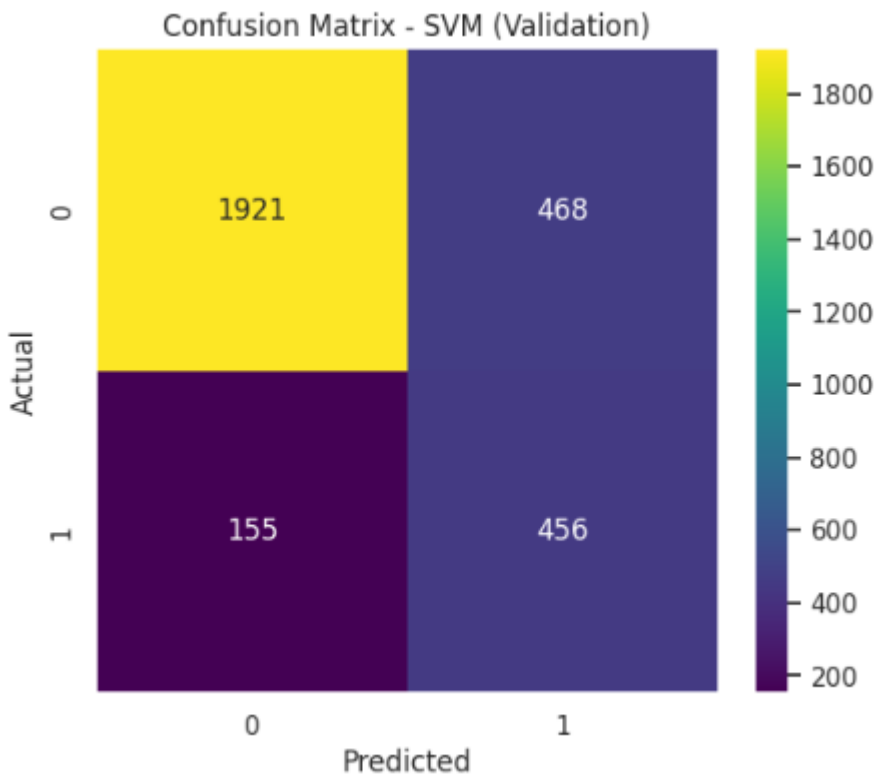
Random Forest



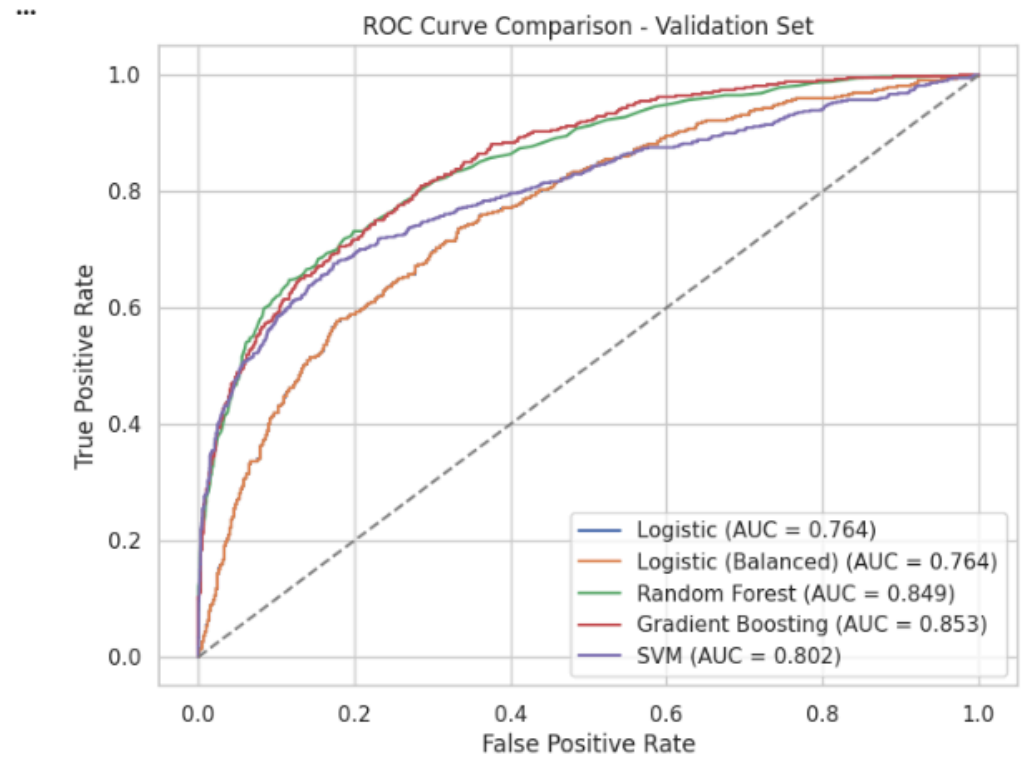
Gradient Boosting


Confusion Matrix - Gradient Boosting (Validation)
2256
133
309
302
Actual
0
1
Predicted
0
1
2250
2000
1750
1500
1250
1000
750
500
250

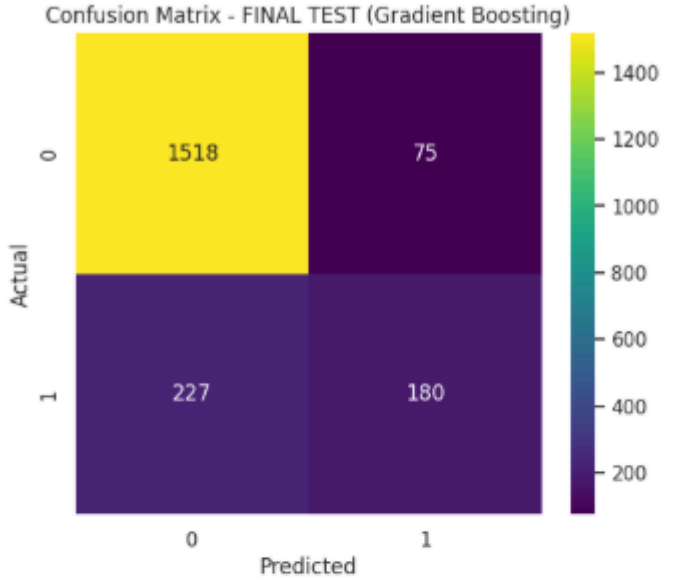
SVM

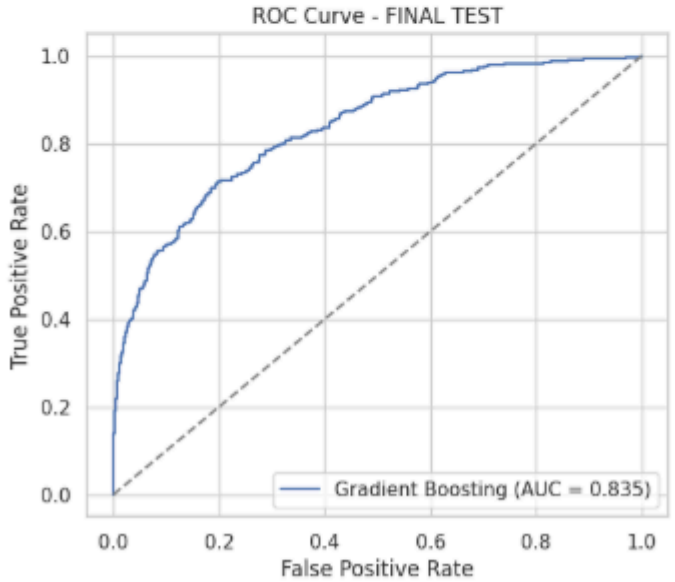

Confusion Matrix - SVM (Validation)
1921
468
155
456
Actual
0
1
Predicted
0
1
1800
1600
1400
1200
1000
800
600
400
200

ROC



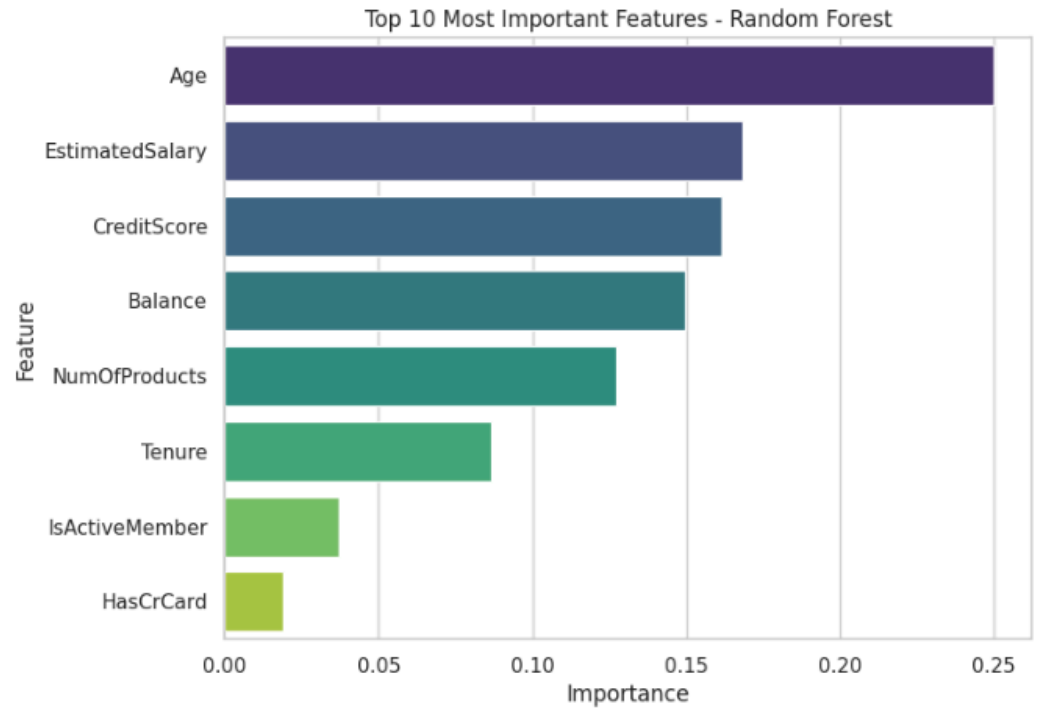
Final Model Selection






```
Final Test ROC-AUC: 0.8351587334638184
```

Feature Importance (RF)



	Feature	Importance
1	Age	0.249862
7	EstimatedSalary	0.168499
0	CreditScore	0.161485
3	Balance	0.149883
4	NumOfProducts	0.127292
2	Tenure	0.086742
6	IsActiveMember	0.037313
5	HasCrCard	0.018925

4. Results

Model Performance – All Features

The following results summarize model performance using all available features.

Logistic Regression

- Best Parameters: {'C': 0.1, 'solver': 'lbfgs'}
- Accuracy: ~0.72
- F1-Score (Churn): ~0.37
- Precision/Recall (Churn): ~0.27 / ~0.62

Logistic Regression achieved relatively strong recall for churned customers, meaning it was able to correctly identify a significant portion of customers who left the bank.

Support Vector Machine (SVM)

- Best Parameters: {'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
- Accuracy: ~0.87
- F1-Score (Churn): ~0.38

SVM achieved high overall accuracy but demonstrated moderate performance in detecting churned customers.

Random Forest

- Best Parameters: {'max_depth': 20, 'min_samples_split': 10, 'n_estimators': 100}
- Accuracy: ~0.87
- F1-Score (Churn): ~0.24

Although Random Forest achieved high accuracy, its ability to detect churned customers was relatively weak.

Gradient Boosting

- Best Parameters: {'learning_rate': 0.1, 'max_depth': 3, 'n_estimators': 200}
- Accuracy: ~0.88
- F1-Score (Churn): ~0.43

- Recall ≈ 180 / (180+227) ≈ 0.44

Gradient Boosting achieved the highest F1-score for churn prediction among all models trained on the full feature set.

High-Correlation Features

When training using only high-correlation features:

- Performance generally decreased.
- F1-score for churn dropped significantly for tree-based models.
- Logistic Regression and SVM maintained moderate performance but did not outperform the full-feature models.

This indicates that limiting the model to only highly correlated features may remove useful predictive information.

5. Discussion

Our models achieved strong overall performance in predicting customer churn, with Gradient Boosting achieving the highest overall performance in terms of ROC-AUC and F1-score for the churn class.

However, despite relatively high accuracy (around 85–88%), all models showed limitations in predicting the minority class (Exited = 1). Since only about 20% of customers churned, the dataset was imbalanced, making churn detection more challenging.

Logistic Regression demonstrated strong recall for churned customers, meaning it identified a higher proportion of customers who actually left. In contrast, Gradient Boosting achieved better balance between precision and recall, resulting in the highest F1-score overall.

Feature Importance

Using Random Forest feature importance, we identified the most influential predictors of churn:

- EstimatedSalary
- CreditScore
- Balance
- Age
- NumOfProducts

These findings suggest that financial indicators and customer engagement variables play a significant role in predicting churn behavior.

Challenges & Improvements

• Class Imbalance

The main challenge was the imbalance in the target variable (Exited = 1 represents approximately 20% of the dataset).
Although we used stratified splitting and evaluated recall and F1-score, further improvements could include:

- SMOTE (oversampling)
- Cost-sensitive learning
- Threshold tuning

• Model Interpretability

While Gradient Boosting achieved the best predictive performance, it is less interpretable compared to Logistic Regression or Random Forest.
For future work, explainable AI methods such as SHAP values could be applied to better understand individual predictions.

• Data Limitations

Although financial features ranked highly, behavioral features such as customer engagement or interaction frequency were limited in the dataset.
Collecting richer behavioral data may improve predictive performance in future iterations.

Future Directions

To further improve the model:

- Apply advanced imbalance handling techniques (e.g., SMOTE)
- Perform threshold optimization for churn detection
- Explore ensemble stacking methods
- Apply explainability techniques (SHAP, LIME)

Final Model Selection

Based on the validation and test results, **Gradient Boosting** was selected as the final model for customer churn prediction.

Among all evaluated models, Gradient Boosting achieved the highest overall performance in terms of **F1-score and ROC-AUC**, while maintaining competitive recall for the churn class (Exited = 1). Although Logistic Regression demonstrated slightly higher recall in some cases, Gradient Boosting provided a better overall balance between precision and recall, resulting in the strongest F1-score.

Given the class imbalance in the dataset (approximately 20% churned customers), evaluating models based solely on accuracy would be misleading. Therefore, emphasis was placed on F1-score and recall for the minority class. Gradient Boosting consistently showed superior discriminatory ability (ROC-AUC ≈ 0.85 on validation and ≈ 0.83 on test data), indicating strong generalization performance.

The final test evaluation confirmed that Gradient Boosting maintained stable performance without significant overfitting, supporting its robustness and suitability for real-world churn prediction scenarios.

For these reasons, Gradient Boosting was chosen as the final model for deployment and further analysis.

📎 Appendix: Use of ChatGPT for Code Assistance

1. Data Loading and Missing Values

Data Loading and Basic Overview

Prompt:
"Help me load a CSV dataset in Python, check the dataset structure (head/shape/info), and understand the basic overview for a churn dataset."

Python

```python
import pandas as pd

# ==============================
# Load the dataset and basic overview
# ==============================

# Load the dataset
df = pd.read_csv("Churn_Modelling.csv")

# Display first rows
print("First 5 rows:")
display(df.head())

# Dataset shape
print("\nDataset shape:", df.shape)

# Data types and non-null counts
print("\nDataset info:")
df.info()
```

2. Summary Statistics + Missing Values

Prompt:
"Show summary statistics for numerical columns and check for missing values in the dataset."

```python
import pandas as pd

# Summary statistics for numerical columns
print("Summary statistics:")
display(df.describe())

# Check for missing values
print("\nMissing values per column:")
display(df.isnull().sum())
```

3.Train / Validation / Test Split (Prevent Data Leakage)

Prompt:
"How should I properly split a churn dataset into training, validation, and test sets before performing EDA to avoid data leakage (using stratified splitting)?"

```python
from sklearn.model_selection import train_test_split

# ====================================================
# Split to Train / Validation / Test (BEFORE EDA)
# ====================================================

df_eda = df.copy()

cols_to_drop = ['RowNumber', 'CustomerId', 'Surname']
for col in cols_to_drop:
    if col in df_eda.columns:
        df_eda.drop(columns=col, inplace=True)

# Split: Train 50%, Validation 30%, Test 20%
df_train, df_temp = train_test_split(
    df_eda,
    test_size=0.5,
    stratify=df_eda['Exited'],
    random_state=42
)

df_val, df_test = train_test_split(
    df_temp,
    test_size=0.4,
    stratify=df_temp['Exited'],
    random_state=42
)

print("Train:", df_train.shape, "Churn:", df_train['Exited'].mean())
print("Val:  ", df_val.shape,   "Churn:", df_val['Exited'].mean())
print("Test: ", df_test.shape,  "Churn:", df_test['Exited'].mean())
```

4. Prompt – Column-Based Churn Exploration

Prompt:

"Start the data analysis by grouping the dataset according to different columns.
For each feature, evaluate how it relates to customer churn, calculate churn rates for different groups or ranges, and present the findings using clear visualizations."

Python

```python
# Required imports
import matplotlib.pyplot as plt
import seaborn as sns

# ==============================
# Age Distribution by Churn Status
# ==============================

# Create a colorful boxplot with improved styling
plt.figure(figsize=(8,6))

sns.boxplot(
    x='Exited',
    y='Age',
    data=df_train,
    palette=["#4CAF50", "#E53935"]  # Green = stayed, Red = churned
)

plt.title("Age Distribution by Churn Status", fontsize=14)
plt.xlabel("Churn (0 = Stayed, 1 = Left)")
plt.ylabel("Age")
plt.show()

# Calculate and display mean age by churn status
mean_age = df_train.groupby('Exited')['Age'].mean()
print("Mean Age by Churn Status:")
display(mean_age)
```

Churn by Age Groups (Train Set Only) 2️⃣ 📦

Python

```python
# Required imports
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# ==============================
# Churn by Age Groups (Train Set Only)
# ==============================

# Work on a copy to avoid modifying original data
df_train_eda = df_train.copy()

# Create age groups
df_train_eda['AgeGroup'] = pd.cut(
    df_train_eda['Age'],
    bins=[18, 30, 40, 50, 60, 100],
    labels=['18-30', '30-40', '40-50', '50-60', '60+']
)

# Calculate churn rate per age group
age_group_churn = df_train_eda.groupby('AgeGroup')['Exited'].mean()

# Plot
plt.figure(figsize=(8,5))
sns.barplot(
    x=age_group_churn.index,
    y=age_group_churn.values,
    palette="magma"
)

plt.title("Churn Rate by Age Group (Training Set)")
plt.ylabel("Churn Rate")
plt.xlabel("Age Group")
plt.show()

age_group_churn
```

Churn by Balance Quartiles (Train Set Only) 3️⃣ 📦

Python

```python
# Required imports
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# ==============================
# Churn by Balance Quartiles (Train Set Only)
# ==============================

df_train_eda = df_train.copy()

df_train_eda["BalanceGroup"] = pd.qcut(
    df_train_eda["Balance"],
    4,
    duplicates="drop"
)

balance_churn = df_train_eda.groupby("BalanceGroup")["Exited"].mean()

plt.figure(figsize=(8,5))
sns.barplot(
    x=balance_churn.index.astype(str),
    y=balance_churn.values,
    palette="viridis"
)

plt.xticks(rotation=45)
plt.title("Churn Rate by Balance Quartile (Training Set)")
plt.ylabel("Churn Rate")
plt.show()

balance_churn
```

5. Prompt – Correlation Matrix of Numerical Features

Prompt:

"Create a correlation matrix for the numerical features in the training dataset.
Calculate the correlations and display them using a heatmap to better understand the relationships between variables and identify strong connections between features."

```python
# ==============================
# Correlation Matrix (Numerical Features)
# ==============================

# Select only numerical columns
numerical_df = df_train.select_dtypes(include=['int64', 'float64'])

# Compute correlation matrix
correlation_matrix = numerical_df.corr()

# Plot heatmap
plt.figure(figsize=(10,8))

sns.heatmap(
    correlation_matrix,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    linewidths=0.5
)

plt.title("Correlation Matrix of Numerical Features", fontsize=14)
plt.show()
```

6.Prompt :

"Create a single evaluation loop that goes over all my models (Logistic Regression, Logistic Regression Balanced, Random Forest, Gradient Boosting, and SVM), prints the classification report for each one, and then plots a visual confusion matrix heatmap for each model on the validation set."

```python
# Required imports
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, roc_auc_score, confusion_matr
import matplotlib.pyplot as plt

# ==============================
# Random Forest Model
# ==============================

best_rf = RandomForestClassifier(
    n_estimators=200,
    random_state=42
)

best_rf.fit(X_train, y_train)

y_val_pred_rf = best_rf.predict(X_val)
y_val_prob_rf = best_rf.predict_proba(X_val)[:, 1]

print("Classification Report (Validation Set - Random Forest):")
print(classification_report(y_val, y_val_pred_rf))

print("ROC-AUC Score:", roc_auc_score(y_val, y_val_prob_rf))

cm_rf = confusion_matrix(y_val, y_val_pred_rf)
disp = ConfusionMatrixDisplay(confusion_matrix=cm_rf)
disp.plot()
plt.title("Confusion Matrix - Random Forest")
plt.show()
```

Feature Importance

Prompt:
"Can you help me plot the feature importance for Random Forest and Gradient Boosting models in order to understand which variables contribute most to churn prediction?"

I used this prompt to structure the feature importance visualization for tree-based models. The code was adapted to my dataset and applied to the trained Random Forest and Gradient Boosting models.
The final plots were generated based on the fitted models in the project and interpreted manually.

Report Writing Assistance
1. Report Structure Assistance
Prompt:
"Help me structure a final report with sections on background, research question, methods, results, and discussion."

2. Report Methods Section
Prompt:
"Can you extend the methods section with a detailed explanation of each step, including data preprocessing, model training, and evaluation?"

3. Results and Discussion

Prompt:
"Help me interpret the results from the classification reports and confusion matrices. I want to explain why certain models performed better and how they handled class imbalance."

Clarifications and Adjustments

- All code suggestions were modified and adjusted to match the churn dataset and the project requirements.
- The final preprocessing pipeline, model configurations, evaluation metrics, and conclusions were implemented independently.
- ChatGPT was used as a technical assistant for guidance and wording support.